

ADITYA S ACHAR

 Bengaluru, India  adityasachar380@gmail.com  +91 9108396979

 [GitHub](#)  [Linkedin](#)  [Portfolio](#)

SUMMARY

Motivated technology professional with hands on experience building practical solutions and working across the development cycle. Experienced in learning new concepts quickly and applying them to real world use cases in structured, team based environment.

TECHNICAL SKILLS

Languages: Python, C, CSS, JavaScript

Databases: MySQL, PostgreSQL, FAISS, Chroma DB

Machine Learning & Deep Learning: Supervised learning & Unsupervised learning, Feature Engineering, Scikit Learn, TensorFlow, PyTorch, Random Forest, LightGBM, XGBoost, Pandas, Numpy

NLP & LLMs: NLP, NLTK, SpaCy, Hugging Face Transformers, RAG, Ollama, Langchain

DevOps, Cloud & Systems: Linux Systems, Windows OS, GCP, AWS, Azure, Version Control (Git, GitHub, GitLab), Docker, Google Vertex AI, Shell Scripting, Jenkins

Tools: JupyterLab, GoogleColab, Node-Red, Postman, PyTesseract, EasyOCR, Openclaw, GHCP CLI

Frameworks: FastAPI, Flask, Rest API

Automation: n8n, Langflow, MCP

INTERNSHIPS AND EXPERIENCE

IT Machine Learning Engineer - Rambus Chip Technologies, Bengaluru **Oct 2025 - Present**

- Built an EDA resource prediction system using **IBM LSF** and **Spectre data**, scaling from **30K to 1.5M+ production jobs** across **4+ years**; evaluated **ML models** and achieved **$R^2 = 0.87$** for **90th-percentile** prediction using **quantile regression**.
- Designed an **enterprise RAG ecosystem** using **BM25**, **vector search**, **reranking**, and **knowledge graphs**, with department-specific knowledge bases delivering **~40% average time savings**; built **GenAI log/document** intelligence workflows using **LLMs**, **OCR**, and **enterprise APIs** for **20–30 finance users**.
- Developed **15 MCP integrations (~20 team-wide)** for **Jira**, **Confluence**, **Jenkins**, and **ServiceNow** using **Python/FastMCP**, supporting **hundreds of users** and **~30% of relevant ticket activity**; **automated NetApp analytics**, VS Code provisioning, and **SFTP-based semiconductor pipelines**.
- Engineered AI developer tooling and agentic workflows across **GitHub Copilot**, **Continue.dev**, terminal agents, and **OpenClaw**, delivering **70+ skills/integrations**, supporting **80–100 daily users** and millions of tokens/day.
- Drove enterprise **AI adoption** through **Level 1/2 training**, enabling engineers to integrate AI-assisted development and **agentic workflows** into daily engineering processes.

Project Engineer Intern - Stovekraft Pvt. Ltd., Bengaluru **Feb 2025 - May 2025**

- Worked with production monitoring **software development**, **ERP systems**, **BOM management**, and manufacturing data tracking in an **OEM production environment**.

Student Intern - Central Manufacturing Technology Institute (CMTI), Bengaluru **Oct 2023 - Nov 2023**

- Gained hands-on experience in Python programming, industrial networking, and **automation systems** used in smart factories.
- Worked with **Linux**, **Node-RED**, **Python scripting**, and **Grafana dashboards** for real-time data visualization and **control system integration**.

EDUCATION

- Bachelor of Engineering** **Oct 2021 - Jun 2025**
St Joseph Engineering College, Mangalore - Electronics and Communication **CGPA - 7.65**
- Pre University Course** **Jun 2019 - Mar 2021**
RN Shetty PU College, Kundapura **Percentage - 80.83**

PROJECTS

Enterprise RAG Platform

- Built an **n8n-based PDF RAG** platform using **ChromaDB**, **Azure OpenAI**, webhooks, and **custom module JavaScript** for evidence-grounded technical document Q&A.
- Implemented **metadata-aware semantic retrieval**, chunk enrichment, deduplication, and citation/page-level visual previews while maintaining strict document boundaries.
- Developed a multimodal **Streamlit RAG** application with **PyMuPDF** for extracting text, tables, captions, and graphics, with configurable chunking, access controls, and persistent **vector/metadata caching**.
- **Designed hybrid FAISS + BM25** retrieval with **reciprocal-rank fusion**, **adjacent-chunk expansion**, and prioritized context assembly; integrated **Azure OpenAI** and **Ollama** for streamed, citation-aware responses.

LSF Spectre Resource Prediction

- Built a modular **Python ML prediction** system to estimate memory and compute requirements for **Spectre simulation** workloads, covering **log parsing**, **data preparation**, **training**, **inference**, and conversational assistance.
- Engineered a pipeline to extract **circuit**, **analysis**, **memory**, **utilization**, and **runtime features** from historical **Spectre logs** and **PostgreSQL** records, generating structured **JSON/CSV datasets**.
- Developed **LightGBM quantile regression models** for **transient**, **DC**, and **AC workloads** with logarithmic transformations, calibration, circuit-size adjustments, and configurable safety margins.
- Built **CPU/core recommendation** and offline inference logic, integrating **Ollama with embeddings**, **cosine-similarity search**, tool invocation, and **contextual natural-language explanations**.

ServiceNow/Jira/Jenkins Assistant for GitHub Copilot Chat

- Built a **Python-based MCP server** connecting **GitHub Copilot Chat** with **ServiceNow/Jira/Jenkins**, enabling **natural-language retrieval** of approved enterprise records directly from **VS Code**.
- Implemented **OAuth 2.0 authentication**, token validation/renewal, HTTPS requests, and runtime configuration checks for secure ServiceNow API access.
- Extended the **MCP-based integration to Jira and Jenkins**, enabling secure API access and tool interactions using **Personal Access Tokens (PATs)**.

Advanced RAG System

- Built an end-to-end RAG system using **Mistral 7B**, **ChromaDB**, and **LangChain** with **hybrid BM25 + semantic retrieval**, **CrossEncoder reranking**, and a **spaCy NLP pipeline**; deployed via **Streamlit** with multi-turn conversational memory.

Portable Soil Fertility Detector using ML

- Portable Soil Fertility Detector is a compact device that measures soil **N, P, K, pH and moisture** to assess fertility, using **sensors** and a **microcontroller** for **real-time analysis**.

FloodCast Using ML and Google API

- Designed and developed a flood prediction and safe route planning system by integrating **weather APIs**, **elevation data**, and **machine learning models**, with an **interactive map interface** built using **React** and **Google Maps**.

ADDITIONAL INFORMATION

- **Certifications:** Google Analytics Certification (GOOGLE), Introduction to programming with MATLAB, Wireless Communication, Simulink and Sim scape Onramp with MATLAB (Mathworks), Basics of Networking (CISCO)
- **Personal Skills:** Leadership, Listening skill, Problem solving, Adaptability, Time Management
- **Hobbies:** Football, Taekwondo, Photography, Video Editing